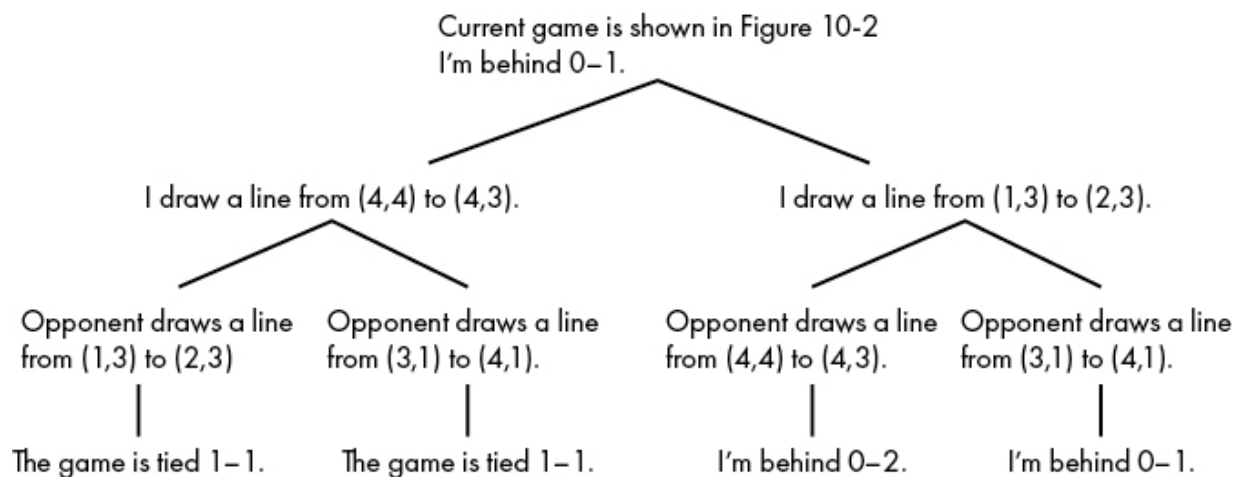


checkers or tic-tac-toe, and an algorithm for winning all those games can give you a new way to think about every competitive situation you encounter in life. The essence of a winning strategy is simply to systematically analyze the future consequences of our current actions, and to choose the action that will lead to the best possible future. This may sound tautological, but the way we accomplish it will rely on careful, systematic analysis; this can take the form of a tree, similar to the trees we constructed in Chapter 9.

Consider the possible future outcomes illustrated in [Figure 10-3](#).



[Figure 10-3](#): A tree of some possible continuations of our game

We start at the top of the tree, considering the current situation: we're behind 0–1 and it's our turn to move. One move we consider is the move in the left branch: drawing a line from (4,4) to (4,3). This move will complete a square and give us one point. No matter what move our opponent makes (see the possibilities listed in the two branches in the bottom left of [Figure 10-3](#)), the game will be tied after our opponent's next move. By contrast, if we use our current turn to draw a line from (1,3) to (2,3), as described in Figure 10-3's right branch, our opponent then has a choice between drawing a line from (4,4) to (4,3) and completing a square and earning a point, or drawing another line like one connecting (3,1) and (4,1), and leaving the score at 0–1.

Considering these possibilities, within two moves the game could be at any of three different scores: 1–1, 0–2, or 0–1. In this tree, it's clear that we should choose the left branch, because every possibility that grows from